

Data Science
Project Proposal
Predictive Modeling and Micro-Climate Mitigation of Urban Heat Islands in Karachi Using GEE and Machine Learning



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1. Problem Definition & Objectives

Rapid urbanization, deforestation, and the rise of concrete structures in Karachi have caused certain areas of the city to trap heat, creating Urban Heat Islands (UHIs).

Objective: This project aims to use Google Earth Engine (GEE) to extract Land Surface Temperature (LST) and other environmental factors. We will then build Machine Learning models to predict how much the temperature could drop if we hypothetically increase vegetation or water bodies in those specific areas. This predictive approach will serve as a foundational "Digital Twin" simulation for Karachi's micro-climate.

2. Dataset

Instead of relying on static CSV files from Kaggle, we will generate a real-time, custom dataset using the **Google Earth Engine (GEE) API** in Python.

- **Data Source:** Landsat 8 or Sentinel-2 satellite imagery.
 - **Features to Extract:**
 - **NDVI:** Normalized Difference Vegetation Index (measures vegetation density).
 - **NDBI:** Normalized Difference Built-up Index (measures concrete/urban structures).
 - **NDWI:** Normalized Difference Water Index (measures water bodies).
 - **Albedo:** Surface reflectivity.
 - **Target Variable:** LST (Land Surface Temperature).
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3. Data Preprocessing & Exploratory Data Analysis (EDA)

- **Cloud Masking:** Using GEE scripts to filter out cloud cover from satellite images for accurate readings.
 - **Data Transformation:** Converting GEE spatial pixel data into a tabular format (CSV/Pandas Data Frame) to train the machine learning models.
 - **EDA:** Generating correlation heatmaps (e.g., proving the positive correlation between LST and NDBI, and the negative correlation between LST and NDVI). We will also provide comparative summary statistics for different towns in Karachi (e.g., DHA, Saddar, Lyari, Gulshan).
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4. Model Implementation

We will apply and compare four different machine learning regression models to predict the target variable (LST):

1. **Random Forest Regressor:** Excellent for capturing non-linear spatial data patterns.
 2. **XGBoost:** A powerful boosting algorithm known for high accuracy.
 3. **Support Vector Regression (SVR):** Effective for generalizing data points on a carefully sampled dataset.
 4. **Multi-Layer Perceptron (MLP) / Deep Neural Network:** Implemented via Keras/PyTorch to bring a deep learning approach to the project.
- **Hyperparameter Tuning:** We will use GridSearchCV or RandomizedSearchCV to find the optimal parameters for all models.
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5. Evaluation

The models will be evaluated using standard regression metrics:

- **Metrics:** Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared (R^2) Score.
 - **Visualizations:** Bar charts and line graphs will be generated to provide a clear comparative analysis of which model performs best.
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6. Novelty & Innovation (Bonus Criteria Addressed)

To demonstrate novelty and real-world application, the project will go beyond basic prediction. We will use our best-performing model to run a "**Mitigation Simulation.**" We will artificially increase the NDVI (vegetation) values by 20% in our test data simulating a hypothetical tree-planting drive in Karachi.

- We will then run the model to predict the new LST and measure exactly how many degrees the temperature would drop. This transforms our project from a simple analysis into an actionable policy-making tool relevant to Climate Change and Urban Heat Island mitigation.
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7. Conclusion

In conclusion, this project addresses the critical and escalating issue of Urban Heat Islands (UHIs) in Karachi by leveraging the analytical power of Google Earth Engine and advanced machine learning techniques. By moving beyond basic exploratory data analysis, our approach integrates spatial data extraction with predictive modeling to create a functional, localized "Digital Twin" of the city's micro-climate.

Through the comparative analysis of four distinct machine learning models (Random Forest, XGBoost, SVR, and MLP), we will identify the most accurate method for predicting Land Surface Temperature based on urban infrastructure and vegetation indices. More importantly, the inclusion of a "Mitigation Simulation" elevates this research from a purely academic exercise to an actionable, data-driven policy tool. Ultimately, this project fulfills all core data science requirements while demonstrating high relevance and innovation in the critical domains of Climate Change mitigation and sustainable urban planning.
